ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION **WINTER SEMESTER, 2022-2023**

DURATION: 1 HOUR 30 MINUTES **FULL MARKS: 75**

CSE 4107: Structured Programming I

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

---

1. a) What are reserved keywords in C? Provide six examples of reserved keywords. 2 + 6 (CO1) (PO1)

   b) Analyze the program mentioned in Code Snippet 1 and show the output. 5 (CO3) (PO2)

```c
#include <stdio.h>

int main(){
    printf("%6d, %4d", 86, 1040);
    printf("%12.5e", 30.253);
    printf("%.4f", 83.162);
    printf("%-6.2g", .0000009979);
    printf("%5.3o", 32);

    return 0;
}
```

Code Snippet 1: Sample program for Question 1.b)

   c) Write a C program that at first takes two integers n and k, and then takes n number of integers sequentially in an array as inputs. The integers are not necessarily given in any order and may be totally random. The program should finally print the three closest integers (having the least difference) to k from the n integers. If there is not enough integers in the array, then it should print "Not enough values". 12 (CO3) (PO1)

2. a) Find the output of the program given in the Code Snippet 2 and provide a proper parenthesization of the expression: 10 (CO3) (PO1)

```c
#include <stdio.h>

int main(){
    int a=1, b=2, c=3, d=4, e=5, f=2;

    a = b += c++ - d + --e / -f;

    printf("%d %d %d %d %d %d", a, b, c, d, e ,f) ;

    return 0;
}
```

Code Snippet 2: Sample program for Question 2.a)

b) Write a C program that takes *100* integers as input, stores those in an array, and prints the total number of duplicate elements in that array. 10 (CO3) (PO1)

c) Explain the use of the break and continue statements in C. 5 (CO2) (PO1)

3. a) Differentiate between implicit and explicit type conversions. 6 (CO2) (PO1)

b) Provide your own justification regarding the length of the array if it is initialized as follows: 5 (CO2) (PO1)

```
int a[] = {4, 9, 1, 8, [0] = 5, 7};
```

c) Find the output of the program given in Code Snippet 3 and justify the results by showing what is evaluated in each of the expressions inside the `if` statements. 14 (CO3) (PO2)

```
1  #include <stdio.h>
2
3  int main(){
4      int a=1, b=20, c=5, d=3;
5
6      if(a>=1 && b>c){
7          printf("This\n");
8      }
9      if(b-- > 19 || c==d){
10         printf("is\n");
11     }
12     if(d<a || ++c == 6 && d<c/2){
13         printf("not\n");
14     }
15     if(!(c<d && a<=d) || b<c){
16         printf("easy\n");
17     }
18
19     return 0;
20 }
```

Code Snippet 3: Sample program for Question 3.c)